

Signaling the Dog: How Profile Content Accelerates Technology-Mediated Dog Rehoming

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Abstract

Peer-to-peer pet rehoming platforms have emerged as a popular alternative to shelter-based placement, enabling owners to find new homes for their pets directly without institutional intermediaries. Despite their scale and reach, little is known about what profile factors drive rehoming speed on these platforms. We analyzed 38,694 dog profiles posted on *Rehome by Adopt a Pet*, a U.S.-based peer-to-peer rehoming platform, to investigate the role of profile features and pet stories on rehoming speed. Using a combination of competing risks survival analysis (Fine-Gray subdistribution hazard model) and Linguistic Inquiry and Word Count 22 approach, we model speed of rehoming as a function of dogs' appearance, categorical profile features, and linguistic features of owner-written pet stories. Results indicate that similar to shelter and rescue contexts, dog appearance dominates peer-to-peer rehoming speed. Compatibility disclosures about dogs' suitability with children, cats, and other dogs significantly accelerate rehoming speed, while the use of risk-oriented language (e.g. *safe*, *protect*, *alert*) in owner-written stories slows it. We interpret these findings within a sociotechnical systems framework, arguing that the *Rehome* platform mediates how "rehome-ability" is constructed and communicated, and that profile story text includes subtle signals of a given dog's "rehomg-ability." We conclude with profile-related recommendations for dog owners and shelter and rescue programs that encourage or require self-rehoming attempts prior to surrender.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Animal-computer interaction, online pet profiles, peer-to-peer rehoming, animal sheltering, dogs, pet analytics, competing risks, big data

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1 Introduction

In 2025, Mocha, "a sweet and loyal 8-year-old" Chihuahua, needed a new home. According to her profile on the peer-to-peer pet rehoming website *Rehome by Adopt-A-Pet*, she was a close companion to her owner "through everything—from hard times to happy ones." The average Chihuahua posted on *Rehome* in 2025 received 3 applications from potential new families and found a new home in just 19 days. But Mocha's profile was on the site for more than four months. She received just one application. Our study is about dogs like Mocha, and what their owners did and didn't say about them on *Rehome* as they sought to find new caregivers for their beloved pets. It is about which profile features were most effective in expediting the process of finding dogs new homes. We wanted to know if Mocha's profile had something to do with her long wait.

Rehoming speed matters. Unlike Mocha's patient owner, 70% of *Rehome* users say they have just four weeks or less to rehome their pet, with 25% of all users reporting they have just a week or less [19]. Users with these time crunches are more likely to remove their dogs' profiles and relinquish them to shelters or rescues than those without [19]. This may be because the majority (59.4%) of owners report needing to rehome their pet because of personal (35.9%) or housing issues (23.5%) [19]. Animal welfare organizations—many of which ask or require owners to seek peer-to-peer rehoming before surrendering their pets—now see *time* as a strategic resource for achieving humane animal outcomes [5].

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Millions of pet dogs like Mocha lost their homes in the United States last year [?]. Animal shelters have long been key actors in matching dogs to new ones, but admission of dogs into these institutions is increasingly seen as disruptive to the human-animal bond and disorienting to animals [12]. The public is more aware than ever that nationally, 1 in 10 dogs surrendered to animal shelters and rescues dies in them [?].

There are alternatives. In 2025, 2 million dogs changed hands through shelters and rescues [?]. However, a similar number also flowed through the private sector via informal peer-to-peer acquisition. When asked where they obtained their new pet, between 27-37% of respondents said they acquired them through non-breeder individuals like friends, family or strangers they met online [? ?]. As social media has increased the reach of the average citizen in networking with others, tech-enabled self-rehoming has emerged to support the transfer of pets like Mocha directly from one family to another.

Growing scholarly interest in how these sociotechnical rehoming systems shape outcomes for pets has focused on their effectiveness for placing them into new homes. A key study of *Rehome* data by Ly and Protopopova (2022) focused on the platform’s success in diverting animals from shelter relinquishment based on animal- and owner-related characteristics [19]. A study by Cornelison et al. (2025) compared rehoming speed of different groups of pets available on *Rehome* to groups of similar animals in 19 U.S. shelters and found that most shelters were able to rehome both easier and harder to adopt animals more quickly [8].

The effect of how users construct profiles for their pets on peer-to-peer rehoming speed is less explored. The effect of owners’ written pet stories on rehoming speed is unknown. Unlike trained shelter staff, individual caretakers vary in their experience with pet profile construction, persuasive writing, marketing, and understanding of animal behavior, leading to variability in the animal profiles they create. This creates a natural experiment for evaluating the influence of free-form story text on rehoming speed.

This study addresses these opportunities using a large dataset of peer-to-peer rehoming profiles from *Rehome* by *Adopt-A-Pet*, a U.S.-based rehoming platform. *Rehome* enables owners to create profiles for pets they need to rehome. Their profiles appear on *AdoptaPet.com* alongside those of shelter and rescue animals. In this paper, we focus on dogs listed on the platform. We ask: (1) What profile features predict rehoming speed? (2) Do linguistic features of profiles’ pet stories improve prediction of rehoming speed and if so, which ones? Stated another way—what platform affordances could Mocha’s family have prioritized, and what could they have written (or not written) in her story to help her find a new home faster?

Our results demonstrate that on the *Rehome* platform, users can make profile construction decisions to accelerate the speed of rehoming success. Answering ‘Yes’ to questions about their dogs’ compatibility with kids, cats, and other dogs instead of skipping these questions increases peer-to-peer dog rehoming speed; so does including certain linguistic features like writing from a family-oriented ‘we’ perspective. Including risk language (e.g. ‘protect’, ‘safe’, ‘alert’) in pets’ stories has the opposite effect, slightly slowing rehoming speed.

We find that the *Rehome* platform’s informational structure mediates how ‘rehome-ability’ is constructed and communicated, and that profile story text includes subtle signals that influence adopter interest. Owners seeking to rehome their dogs may therefore benefit from employing a persuasive profile creation strategy similar to that used by shelters or rescues. Such a strategy would also benefit shelters and rescues that increasingly rely on owner-self rehoming to reduce the flow of animals coming through their doors.

2 Related Work

Our work is situated within on-going scholarly conversations about sociotechnical systems for animal management, platform affordances in peer-to-peer pet rehoming contexts, and pet selection decision-making, i.e., dog adopters’ preferences for certain kinds of animals.

2.1 Rehoming Platforms as Sociotechnical Systems

Rehome operates as a sociotechnical system. Its interface design and information architecture shape what information is collected from pet owners, how it is displayed, and how users browsing for a new pet use that information for search and decision-making. Digital classification systems like *Rehome* are never neutral—they encode values, assumptions, and power dynamics, and have real material effects on the things and people being classified [?]. The animal welfare literature supplies some empirical evidence for these effects. Gunter et al. (2018) genetically tested 919 shelter dogs and found that visual breed identification by shelter staff matched DNA results for at least one breed in 67.7% of dogs, but fell to 10.4% in identifying more than one breed. Staff were more accurate in identifying dogs with more pit bull heritage, creating information asymmetry in how the breed classifications were applied. Pit bull-type dogs already suffer prejudice in much of the U.S. and are a common target of breed-restrictive housing practices and legislation [1]. Cohen et al. (2020) demonstrated the effects of removing breed labels in the shelter setting [6]. Median length of stay dropped by 37%, and the return rate stayed constant at 6%, suggesting the labels caused dogs to wait longer for new homes [6].

What dog characteristics are prioritized as informational elements in rehoming contexts—and which garner the most user-engagement—is a design choice that makes pet rehoming platforms socio-technical rather than simply technical. Unlike dogs in animal shelters who can practice a limited form of agency and self-advocacy beyond institutional classification systems because they can demonstrate kennel presence and personality in view of potential adopters, dogs whose profiles are placed online for peer-to-peer rehoming compete for adopter attention entirely through human-mediated communication. Platform affordances and decisions made by owners when interacting with those affordances serving as their advocacy.

One goal of this study is to identify which classifiers are important to people who shop for adoptable dogs online and to identify which ones owners seeking to rehome their dogs could engage with for better advocacy of their pets.

2.2 Platform Affordances in Pet Adoption Platforms

Like users of human dating platforms, *Rehome* users who post their dogs primarily construct profiles by selecting dogs' features from a series of radio buttons and dropdown menus. Many of these features like age, sex, breed, coat color, and vaccination and spay/neuter status are required, while others are optional. Users also can upload up to four photos and write a pet story up to 1,000 characters in length. Users can edit their story after it is published to go beyond this limit. On the other side of the matching equation, many of these features play a prominent role in the potential adopter experience. The primary UI for viewing available dogs is a grid layout with up to a dozen short-form profile cards featuring a dog's photo, name, breed label, sex, age, and location. Prominent filtering tools allow users to search for dogs by breed, age, sex, size, 'good with' markers (compatibility with kids, cats, and dogs), coat color, spay/neuter status, special needs, and vaccine status as well as additional markers like newness to the site and house training status. These features are largely similar on other peer-to-peer and institutional (shelter and rescue) pet acquisition platforms.

2.3 Adoption and Rehoming Decision-Making

Most research into how potential adopters select new dogs has been conducted in shelter environments. Studies of shelter pet adopter preferences have concluded that no single factor can determine adoption outcomes of pets, due to the wide variety of adopter preferences for different kinds of animals [10, 11, 15]. Even so, some factors tend to dominate decision-making.

2.3.1 Appearance Factors. Visitors to shelters tend to prioritize morphological features of pets such as age, size, and health status when evaluating them for adoption [3, 10, 13]. Age effects are robust across both dogs and cats, with younger animals typically adopted more quickly [25, 26, 30]. Appearance is frequently cited by adopters of puppies and kittens as a primary driver of adoption decisions [29], and sits among the top three characteristics potential adopters of all ages of dogs said they considered prior to adoption [10]. Physical attributes such as coat color, size, and breed have repeatedly been associated with differences in length of stay for dogs at animal shelters [25] [9]. Researchers using Petfinder Malaysia data found that they could often predict adoption speed of shelter and rescue animals on the site using only categorical profile features like breed, age, and spay/neuter status [31].

These characteristics have also been found to affect rehoming speed for dogs on online rehoming platforms. A comparative study of *Rehome* data and data from 19 animal shelters that focused on large dogs, senior dogs, and senior cats found that this group of 'hard to adopt' animals spent longer awaiting rehoming in both shelters and on *Rehome* than their younger, smaller counterparts [8]. These findings underscore the potentially outsized importance of profile features related to dog appearance in online dog rehoming speed and outcomes.

2.3.2 Pre-Set Personality and Behavioral Factors. Social behavior toward adopters and perceived personality also have been among the most frequently cited factors for how adopters selected a pet in shelter settings [22, 27, 29]. In one study of in-shelter dog adoption

preferences, adopters reported that appearance, train-ability, and compatibility were the top three characteristics they considered *prior* to adoption but personality was, on average, the factor most highly cited *after* they selected a pet for adoption [10].

The average person is about as good at identifying personality characteristics in dogs as they are in other humans [?], and minimally-trained laypersons have demonstrated capability in rating dog behavior [?]. However, known biases may affect potential adopter's assessments of the personality and behavioral characteristics of adoptable dogs. Owner personality biases dog personality ratings [?]. More conscientious owners rate their dogs higher on training focus and amicability personality dimensions, while more extroverted owners thought their dogs were more amicable [?]. In one study of shelter dog adopters, Cohen and Todd (2019) compared adopters' personality and behavior ratings of their chosen dogs to results from the Match-Up II behavior evaluation tool performed on the dogs by shelter staff. They found little relationship between the two measures, suggesting that adopters may hold different concepts of personality and behavior dimensions than professionals, or lack the ability to accurately assess them in adoptable dogs [7].

Personality and behavioral factors on the *Rehome* platform are prioritized in profile and filtering UIs through the 'good with dogs', 'good with kids', and 'good with cats' features, which users can mark as either 'Yes', 'No', or 'Unknown.' Dogs that are identified as good with other dogs, kids, and cats have increased odds of being diverted from shelter relinquishment (either rehomed or kept by their owners) on *Rehome* [19]. »Add bit about kept/relinquished« *Rehome* also provides a list of personality tags like 'affectionate,' 'snuggly,' and 'protective' that users can add to their dog's profiles. However, personality tags are *not* part of initial profile construction; instead, they can be added in a profile editing flow after an animal's profile has been published. Personality and behavior descriptions may also emerge in owner-written pet stories.

2.3.3 User-generated content. Despite the dominance of appearance and personality markers in adoption decision-making, user-generated profile listing features like the content and number of photos and linguistic features of pet stories have been found to be associated with adoption speed in both in-person and online shelter and rescue contexts.

Photos For online adoptable dog profiles, photo features like overall photo quality, use of outdoor locations, eye-contact from the dog in the picture [17], and closed mouths, floppy ears, and the use of indoor kennel locations have been associated with faster adoption speed [23]. *Rehome* supports the addition of zero to four photos to caretaker-created profiles, with more photos associated with increased odds of diversion from shelter relinquishment on the platform, compared to having no photographs [19].

Pet Stories Pet stories have been assumed by shelter and rescue organizations to serve as a powerful layer of persuasion within adoption profiles. National animal welfare organizations like the ASPCA and Best Friends Animal Society emphasize the role of pet stories as persuasive tools in instructional guides aimed at improving shelter and rescue workers' pet marketing skills [ASPCA][BFAS]. Persuasion is a well-established construct within communication research, encompassing linguistic, emotional, and narrative mechanisms that can shape attitudes, perceptions, and behavior across

media contexts [14, 28]. Research in other pro-social domains—including environmental communication and health messaging—demonstrates that subtle linguistic features (e.g., emotional tone and narrative perspective) can meaningfully affect user engagement and decision-making [4, 20?]. User-written Community Notes on the social media platform X that contextualize animal behavior have been demonstrated to decrease negative user attitudes toward wildlife, emphasizing the power of text in persuasion around animals online [?].

Within animal adoption research, no studies have explicitly examined the linguistic qualities of profile text on peer-to-peer rehoming platforms. Some work suggests that language choice, framing, and narrative style may influence adoption outcomes for shelter and rescue pets with online profiles [16]. Research by Markowitz (2020) using the Linguistic Inquiry and Word Count (LIWC) inventory has shown that shelter and rescue pets whose online profiles demonstrated a more analytic thinking style and relied less on social words were adopted faster [21]. Research into the use of adjectives in dog story text on the Australian online pet adoption platform PetRescue.com found that profiles featuring the adjective phrases, “make you proud,” “independent,” “lively,” “eager,” and “clever” were associated with shorter times to adoption, although the effect was different across the four breed groups studied [24]. Stories with words like ‘affectionate,’ ‘calm,’ ‘eager,’ ‘friendly,’ and similar personality words were found to improve the perceived adoptability of dogs in an experimental design around shelter dog profile features, whereas the word ‘energetic’ reduced perceived adoptability [2]. A study of pronoun use in shelter cat descriptions found a small, significant increase in adoption speed for cats whose online biographies were written in third-person compared to first or second [26].

In the same study of *Petfinder* Malaysia data that looked at online pet adoption factors and produced effective adoption predictions using only morphologically-based profile features like breed, age, and spay/neuter status, two of the five models tested performed slightly better with the inclusion of free-form text, suggesting that text may play a small role [31]. Importantly, the authors noted that text descriptions were generally homogeneous in their word choice, and failed to produce clearly delineated topical differences through topic modeling. The authors identified the homogeneity of pet story language as a key issue in assessing the role of textual factors in adoption speed [31].

» Need transition here«

3 Data Collection

Data for this study were obtained from *Adopt a Pet*, and consisted of more than 111,000 dog adoption profiles posted to the *Rehome* by *Adopt a Pet* platform by pet owners from Jan. 13 to Sept 9, 2025. The dataset represents individual dogs (each identified by a unique `pet_id`) that had active profiles on the platform across the United States during the 2025 calendar year, with a small number (410) posted by residents of Canada.

3.1 Inclusion Criteria

To delist their dogs from the *Rehome* platform, dog owners must select a reason from a dropdown menu. The dataset was filtered

to only profiles that were either still active on the site when the dataset was assembled at the end of 2025 (censored, $n = 10,716$) or that had user-reported reasons for removal indicating the dog was: (1) rehomed to a peer via the *Rehome* platform (rehomed_p2p, $n = 13,941$), (2) kept by the owner (kept, $n = 9,200$) or (3) relinquished to a shelter or rescue (relinquished, $n = 4,837$). Profiles with outcomes indicating the dog was rehomed to a friend or ‘other’ source were excluded from the dataset as there is no way to know what role the dog’s online profile played in those rehoming events. A dog owner who posted their dog on the platform may have shared a link to their dog’s profile on Facebook, for example, and found an adopter there. In that case, they may have marked the rehoming event as occurring outside of the peer-to-peer platform even though the profile played an active role in it. An exploratory analysis including off-site adoptions as a fourth competing risk yielded poorer model performance, suggesting that the inclusion of off-site adoptions introduces noise that reduces model discrimination.

Rehome profiles expire after 30 days, and users must either mark an outcome for their pet by the end of each 30-day period or login to reactivate their pet’s profile. This platform requirement ensures that dogs who spent a long time awaiting rehoming were still actively looking; Mocha’s owner would have had to login four times after initially creating her profile to keep her active on the site. About 90,000 profiles that expired were excluded from the dataset. Profiles that were unpublished drafts at the time of data collection also were excluded. The resulting dataset consisted of 38,694 dog profiles with known outcomes.

3.1.1 Rehoming Speed. Rehoming speed was operationalized as the number of days a dog’s profile remained active on the *Rehome* platform prior to either successful on-platform rehoming or profile removal due to owner-retention or dog relinquishment, captured in the dataset with the variable `days_on_site_final`. This continuous measure reflects the duration between listing creation and outcome resolution, providing a more granular representation of rehoming dynamics than adoption outcome alone. For profiles still active at the time of data collection, the number of days on site was calculated using the date of data retrieval as an end date.

4 Methods

4.1 Competing Risks Modeling

Studies of adoption or rehoming speed have usually used Kaplan-Meier survival analysis and standard Cox proportional hazards models, defining the event of interest as adoption and treating all other outcomes as censored events [8? ?]. But if a dog’s online profile was removed because they were euthanized or relinquished to a shelter that doesn’t mean the dog was not adoptable or rehome-able. On *Rehome*, rehoming and relinquishment are competing outcomes which may be driven by outside factors such as relinquishment waiting times and owner life events. Oyama et al. (2018) warn that treating such competing events as censoring in Kaplan-Meier or Cox analysis overestimates the cumulative incidence of the event of interest [?], in this case rehoming.

The Fine-Gray subdistribution hazard model directly estimates the cumulative incidence function (CIF)—the actual probability of a specific outcome over time, accounting for the fact that other

outcomes remove animals from the population of interest [?]. This approach can estimate the real-world probability that a dog with certain characteristics is rehomed by a specific day (e.g. day 30), given that some dogs will be relinquished or kept before then.

A parallel from foster care research reinforces this approach: McDonald et al. (2007) analyzed permanent placement outcomes for 25,000 children in foster care and concluded that different outcomes (reunification, adoption, guardianship) should be analyzed as competing risks because grouping them produces misleading results [?]. The analogy to dogs on *Rehome* who have multiple exit pathways is direct. Fine-Gray competing risks analysis also enables multivariate assessment of the independent contribution of each feature class to the rehoming outcome. This approach is well-suited to the right-skewed structure of the rehoming speed variable and our interest in evaluating profile features' effects on rehoming speed.

Accordingly, our study used a Fine-Gray competing risks analysis to examine the relationship between profile features and rehoming speed, while accounting for the competing risks of users keeping or relinquishing their dogs. Fine-Gray modeling was conducted in R Studio using the *survival*, *ggsurvfit*, *tidycmprsk*, and *gtsummary* libraries and specified without penalization [?]. Concordance indices were computed to assess overall model discrimination. Cox proportional hazards regression was also conducted in R Studio as a sensitivity check; coefficients were similar across both approaches although the Fine-Gray model demonstrated more discrimination power.

4.2 Feature Engineering

Three primary feature classes were included in our competing risks analysis: appearance descriptors known to be key predictors of adoption and rehoming speed (breed, age, size, purebred status); pre-set personality, behavior, and care characteristics entered by users with radio buttons and dropdowns; and linguistic features derived from owner-written pet stories.

4.2.1 Appearance Descriptors. Most profile features are displayed on *Rehome* and represented in the dataset as categorical data, and this is true for dog characteristics relating to appearance. 'Age' is stored as puppy, young, adult, senior and 'Size' is stored as small, medium, large, and x-large. Dog 'Age' and 'Size' were encoded on a four-point scale reflecting known adopter preferences for small and young dogs (and puppy = 2; young = 1, adult = 0, senior = -1 and small = 2; medium = 1, large = 0, x-large = -1).

The dataset included 308 breed identifiers which were organized by one author into breed groups according to the American Kennel Club's 2025 breed group list, resulting in the breed group categories Herding, Hound, Toy, Non-Sporting, Sporting, Terrier, Working, Foundation Stock Service (FFS), and Miscellaneous Class [?]. Mixed Breed (Large), Mixed Breed (Medium), Mixed Breed (Small), and Mutt breed identifiers were merged into a Mixed Breed ($n = 1,717$) breed group. The breed American Pit Bull Terrier ($n = 2,764$) was retained as a breed group as it is not recognized by the AKC but has been of interest in other breed-based adoption research. Mixed breeds identified with colloquial breed names like goldendoodle, chiweenie, and maltipoo were merged into a Designer breed group ($n = 3,936$). 11 breeds ($n = X$) not recognized by the AKC were

matched with breed groups according to their historical use and other breeds with which they shared genetic heritage. The Terrier Breed group was left out of our model to serve as a reference group; breed group results therefore represent rehoming speed compared to dogs in the Terrier group.

Purbred status is entered by users as 'Yes,' 'No,' and 'Unknown'. For analysis, Purebred status was encoded on a three-point scale according to known adopter preferences for purebred dogs, and reflecting confirmed presence of the quality (Yes = 1), confirmed absence (No = -1), and unknown status (Unknown = 0).

Coat color, made up of 29 overlapping descriptive categories (e.g. 'Black', 'Black with Tan, Yellow, or Fawn', 'Tan, Yellow, or Fawn with Black', 'Black with White', 'White with Black') was excluded from the dataset in order to simplify the model and avoid collinearity issues.

My details



Figure 1: Caption

4.2.2 Pre-set personality, Behavior, and Care Characteristics. The remaining profile features are structured, categorical data collected via radio buttons and related to dogs' personality, behavior, and care characteristics (e.g. special needs status). Users enter this information as either Yes, No, and Unknown. For analysis, all Yes/No/Unknown profile fields were encoded on a three-point scale reflecting confirmed presence of the quality (Yes = 1), confirmed absence (No = -1), and unknown or unspecified status (Unknown = 0). Because unknown statuses typically result in the associated characteristic not being shown on a dog's profile at all, this encoding distinguishes between profiles that actively displayed a characteristic's presence or absence and those that left it unaddressed. As shown in 1, answering 'Yes' to the 'Good With Dogs' category places a green check mark and the words 'Good with dogs' on a dog's profile; answering 'No' places a red caution sign and the words 'Not Good With Dogs'; while answering 'Unknown' doesn't place any reference to the category on the dog's user-facing profile. This encoding reflects known preferences for dogs that are compatible with other dogs, kids, and cats [19].

The amount of user-generated content included in each profile was operationalized as photo counts (the number of photos uploaded by users from 0-4), description character lengths (from 0 to 15xx), and presence of personality trait tags. The vast majority of profiles featured both photos and descriptions, as shown in Table X. Only X profiles lacked photos and only X lacked a description. Whether a profile featured personality trait tags was encoded with 1 representing presence and 0 representing absence. XX% of profiles features personality trait tags. These continuous measures of photo and text volume and presence of personality trait tags enabled the analytic models to distinguish effects of description presence and length from effects of specific linguistic content. Photos were not available in the dataset; qualitative measures of photo quality or content were excluded from the analysis.

4.2.3 Linguistic Features of Pet Stories. Descriptions were cleaned using the Python library BeautifulSoup to strip HTML tags and extract plain text. Common HTML entity codes (e.g. ’, ‘) were replaced with their plain-text equivalents. Informal language like *w/* for *with* was normalized and double hyphens were spaced out. Additionally, bullet pointed lists were converted into sentences and whitespace was normalized.

Cleaned descriptions were analyzed using the full 119-item Linguistic Inquiry and Word Count (LIWC) inventory in the LIWC-22 software. The TextStat Python library was used to calculate a Flesch–Kincaid grade level for each description.

LIWC is a text analysis inventory that counts words in psychologically meaningful categories against an expert-curated dictionary [?]. It is a transparent, dictionary-based approach that maps text to dimensions like attentional focus, emotionality, social relationships, thinking styles, and individual differences. LIWC categories can reflect the underlying psychology of user-generated text, making it a useful tool for profile text where the writer’s psychological stance may influence outcomes [?]. LIWC also provides standardized, reproducible scores across texts and requires no training data. Van der Lee et al. (2019) compared LIWC against machine learning classifiers and human coders for analyzing online dating profiles and found that LIWC’s content-related labels “corresponded more strongly to humans than those of the classifier” while all three approaches performed similarly on the example prediction task [?].

Some limitations, though, are well-characterized. Franklin (2015) highlights that LIWC is fundamentally a “context-independent, word-counting approach” and that it cannot detect sarcasm, negation, or pragmatic meaning [?]. A sentence like “this dog is not aggressive at all” would register “aggressive” in LIWC’s negative emotion categories despite conveying reassurance.

» Qualitative Method here«. Exploratory modeling using keywords generated from content analysis failed to improve the model and were not included in our final analysis, likely because of the wide variety of synonyms used to describe the content topics and overlap with similar LIWC categories like health, time, lifestyle, need, want, and pronoun categories.

4.3 Dimension Reduction

Prior to modeling, dimensions were reduced in the LIWC dataset. Metadata features (Segment, WPS, Dic, etc.) and aggregate parent features that are composites of their subfeatures (e.g. Cognition, Social, Affect, Drives) were dropped to avoid collinearity issues. Parts of speech (verb, adj, prep, det, auxverb, conj, adverb, article, number, quantity) and low-prevalence categories appearing in less than 5% of listings (e.g. fatigue, sexual, relig, politic) were removed to simplify the model.

A collinearity analysis also was conducted in Python using the Statsmodels library to identify and address problematic inter-correlations among features. Variance inflation factors (VIF) and pairwise Pearson correlations were examined across all candidate predictors. Features exhibiting correlations exceeding $r = 0.80$ and $VIF > 10$ with other predictors were flagged for review, and redundant features were removed or consolidated prior to final model

specification. This resulted in the removal of 9 model features, including whether the profile had a description, the Flesch–Kincaid grade level score, and seven LIWC categories (e.g. male, female, and shehe, which correlated highly with the profile’s male/female sex identifier).

4.4 Model Iteration

To assess the incremental predictive contributions of the three profile features sets of interest (appearance descriptors; pre-set personality, behavior, and care characteristics; and linguistic features of pet stories) the model was estimated using a sequential block entry design. Variables were entered in blocks according to feature category (Block 1: breed, age, size, purebred status; Block 2: adds categorical markers and user-generated content counts, and Block 3: adds linguistic text features), with each block tested for significant improvement in model fit relative to the preceding specification using the likelihood ratio chi-square statistic. The significance of Block 3’s contribution to model fit is the primary test of whether listing language predicts rehoming speed over and above what is explained by appearance descriptors and categorical characteristics alone.

4.5 Statement on Researcher Positionality

The lead author of this study brings direct practitioner experience in animal shelter adoption to this research, which was essential in encoding profile features to match common adopter preferences and interpretation of model results. From 2023 to present, she has served more than 1,000 volunteer and work hours at a large, open-admission municipal animal shelter, serving as a paid employee in an Adoptions Counselor role from May to September 2024.

5 Findings

5.1 Profiles Focused on Personality, Appearance, and Behavior

Most dogs profiles identified them as not purebred (61%) and not special needs (90%), spayed/neutered (66%), up-to-date on shots (79%), and house-trained (85%), as shown in Figure 2. Fewer than 6% of users did not provide inputs to radio buttons and drop down menus about these attributes. Most dogs were identified as good with dogs (77%) and kids (77%), and 37% were identified as good with cats. 45% of profile creators did not address their dog’s compatibility with cats. 16% did not address compatibility with kids and 11% did not address compatibility with other dogs. Table ?? shows that most profiles had 3 or 4 photos. Most (96.7%) featured written pet stories and their average length was 561 characters. 65.9% of profiles featured personality trait tags.

Table 1 describes the breed identifiers of the sample. Toy (12.8%), Herding (16.1%), Working (14.6%), Designer (10.2%), and Sporting (11.6%) breeds were about equally represented in the dataset. Terrier (7.7%), Non-Sporting (6.1%), Hound, (5.8%), and Pit Bull Terrier (7.1%), Mixed Breed (4.4%), Foundation (3.0%), and Miscellaneous Class (0.4%) breeds made up the rest of the dataset.

Results of qualitative content analysis are shown in Table 2. The majority of profiles repeated the kinds of information available in the platform’s appearance and pre-set personality, behavior, and

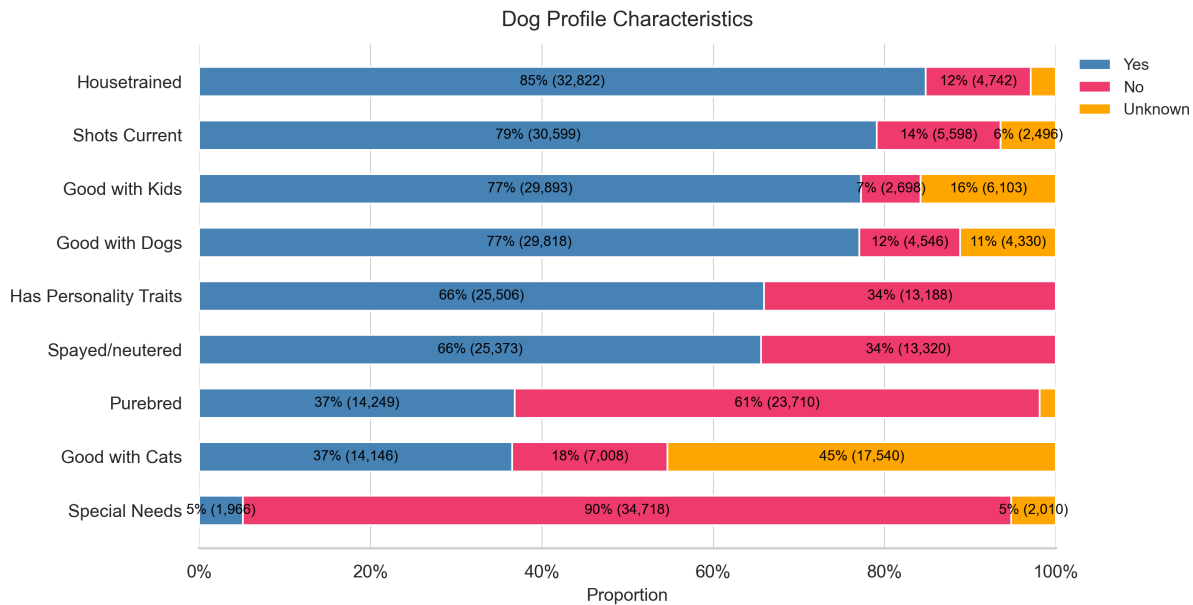


Figure 2: Proportion of dogs by profile characteristics. Most dogs were marked on their profiles as housetrained, up-to-date on vaccines, good with kids and dogs, and spayed or neutered. Most had personality traits listed on their profile. Most dogs were not marked as purebred, good with cats, or special needs.

Table 1: Dog Listings by Breed Group

Breed Group	Count	%
Herding	6,244	16.1
Working	5,652	14.6
Toy	4,966	12.8
Sporting	4,499	11.6
Designer	3,936	10.2
Terrier	2,995	7.7
Pit Bull Terrier	2,764	7.1
Non-Sporting	2,362	6.1
Hound	2,246	5.8
Mixed Breed	1,717	4.4
Foundation	1,173	3.0
Misc	140	0.4
Total	38,694	

care features. Popular topics included a dog’s personality (87.9%), physical attributes like age, size, breed, or color (78.8%), training (63.6%), social experiences or preferences with other dogs, cats, or people (54.5%), placement history (54.5%), and reason for rehoming (54.4%). Most profiles had language centering the dog’s needs, likes, or preferences (69.7%), while few took an angle of describing the dog as deserving of a new home (18%) or used humor (3.0%). Just under half included language in which the profile authors centered their own needs, likes, or preferences in their rehoming search (48.5%) and very few made appeals to profile readers directly (6.1%).

95% of profiles were identified by LIWC software as using a positive tone and having language with a focus on the present. 86% used positive emotional language. About one in five profiles (19%) featured *risk* language. 53% used at least one *I* pronoun, 45% used the first-person pronoun *We*, 37% used *You*, and 25% used *They*. 90% used *she* or *he* pronouns.

5.2 Most Dogs Were Successfully Rehomed

While the majority of dogs (62%) placed on the platform were rehomed, the percent of cases that experienced the Rehomed event (46%, $n = 13,941$) was roughly equivalent to the percent that experienced competing outcomes ($n = 14,037$) and the censored group ($n = 10,716$) in the analyzed dataset, due to the removal of outcomes that were not verified by users to have occurred through the *Rehome* platform. The cumulative incidence of these competing outcomes is shown in Figure 3, showing that most outcomes of any type occur within the first 30 days and follow similar trajectories. By day 90, the vast majority (82%) of dogs were rehomed, kept, or relinquished. The median number of days profiles were on the site for each outcome is roughly equivalent, with Rehomed dogs spending a median of 12 days on the site, relinquished dogs spending 14 days, and kept dogs spending a median of 11 days on the site. Interestingly, many owners who posted profiles to *Rehome* decided to keep their pet within the first few days, with 20% of profiles in the dataset removed by day 30 because of owner retention.

Table 2: Percentage of Rehome dog profiles including each topic at least once ($n = 33$)

Topic	Example	Profiles (%)
<i>Topics</i>		
Personality	<i>e.g., Her personality is sweet, curious, loyal, alert, playful, and very submissive.</i>	87.9%
Physical attributes	<i>e.g., Hector is about 5 mos old and is a blue American Pitbull Terrier.</i>	78.8%
Training	<i>e.g., He is house trained and kennel trained.</i>	63.6%
Social experience / preferences	<i>e.g., Riley is very sweet and good with kids and animals when introduced properly.</i>	54.5%
Placement history	<i>e.g., She was found as a stray a few months ago.</i>	54.5%
Reason for rehoming	<i>e.g., We feel it's best for her to find a home that doesn't cause her as much stress as ours does.</i>	54.4%
Health / medical	<i>e.g., Pita is a healthy dog, sleeps well, eats well, no sicknesses of any sort.</i>	42.4%
Ideal home characteristics	<i>e.g., He needs a high fence and jumps over a standard chain link.</i>	36%
How to adopt / next steps	<i>e.g., If you're interested in more information, message.</i>	15.2%
Supplies dog comes with	<i>e.g., He's on a salmon based diet with allergy treats daily (I will send him with a new bag and a month supply of treats).</i>	6.1%
<i>Perspectives</i>		
Centers animal's needs/likes/preferences	<i>e.g., She is a very sweet dog who just needs a calm home and lots of love.</i>	69.7%
Centers profile author's needs/likes/preferences	<i>e.g., I am looking for someone that has time for him.</i>	48.5%
Centers reader's needs/likes/preferences	<i>e.g., If you're looking for a sweet, steady, and loyal best friend, Gracie is your girl!</i>	6.1%
<i>Approaches</i>		
Animal deserves / has earned	<i>e.g., We are looking for a more loving environment where he can get all the pets, love and attention he needs and deserves.</i>	18.2%
Humorous / playful	<i>e.g., Also a funny thing about Kash is that he farts a lot, he is just an old man that toots.</i>	3.0%

5.3 Model Discrimination

As discussed in section 4.4, a competing risks model was estimated using a sequential block entry design to understand the contributions of the three main kinds of profile content: (Block 1) appearance data, (Block 2) pre-set personality, behavior, and care data, and (Block 3) linguistic features of pet stories.

Block 1 yielded a concordance index of 0.703. Adding pre-set personality, behavior, and care characteristics in Block 2 improved concordance to 0.727 (+0.024). The addition of LIWC-derived pet story features in Block 3 produced a further small improvement to 0.731 (+0.004), confirming that listing language contributes to rehoming speed prediction over and above appearance and categorical features, albeit modestly.

Likelihood ratio tests confirmed that each block contributed significantly to model fit. The addition of categorical features in Block 2 produced a significant improvement over the structural baseline ($LR = 1253.78$, $df = 9$, $p < .001$). The addition of LIWC-derived pet story features in Block 3 produced a further significant improvement over Block 2 ($LR = 244.52$, $df = 57$, $p < .001$).

5.4 Appearance Dominates

Model contributions of each statistically significant profile feature are shown in ???. Table 5 further describes the cumulative incidences of rehoming by day 90 for each feature. Breed was a strong contributor to rehoming speed, with Designer dogs ($HR = 1.85$, $p < .001$) being rehomed at almost 1.9 times the rate of the reference category, Terriers. Toy breeds ($HR = 1.68$, $p < .001$) at about 1.7 the rate of Terriers. Mixed Breed dogs ($HR = 0.58$, $p < .001$), Pit Bull Terriers ($HR = 0.55$, $p < .001$), and Foundation breeds ($HR = 0.64$, $p < .001$) were associated with slower rehoming speeds, with Mixed Breed dogs rehoming just above half (58%) the rate of Terriers. By day 90, 60% of Designer and 59% of Toy breeds were rehomed, compared to just 21% of Mixed breed dogs and 29% of Pit Bulls.

Dog size was a strong predictor ($HR = 1.32$, $p < .001$), with smaller size associated with faster rehoming. Age showed a similarly strong positive association ($HR = 1.21$, $p < .001$), with younger dogs rehoming faster and puppies the fastest.

Purebred status was also positively associated with rehoming speed ($HR = 1.24$, $p < .001$). Special needs designation was negatively associated with adoption speed ($HR = 0.81$, $p < .001$). 27% of dogs

labeled as special needs were rehomed by day 90, compared to 43% labeled as not special needs.

Notably, despite the major contribution of these features to rehoming speed, the percent of dogs kept across all feature categories by day 90 hovers around 25-30% even for these highly salient categories. This suggests that whether an owner kept their dog was not driven by the same factors as rehoming.

5.5 Pre-set personality, Behavior, and Care Features Contribute To Rehoming Speed

The compatibility profile fields ‘good with dogs’, ‘good with kids’, and ‘good with cats’ were all positively associated with faster rehoming.

By day 90, 46% of dogs labeled as ‘good with dogs’ had been rehomed, compared to 22% of dogs labeled as not good with dogs and 34% whose profiles did not display either marker and was overall associated with faster rehoming ($HR = 1.28, p > .001$). ‘Good with kids’ ($HR = 1.28, p > .001$) showed similar results, with 46% of dogs marked as ‘good with kids’ rehomed by day 90, compared to 29% labeled not good with kids and 26% without either marker. Interestingly, dogs without a marker of compatibility with kids rehomed more like dogs with a ‘not good with kids’ marker. 48% of dogs marked ‘good with cats’ ($HR = 1.18, p > .001$). rehomed by day 90 compared to 27% marked as not good with kids and 43% without either marker, suggesting dogs for whom ‘good with cats’ status was unaddressed rehomed more like dogs with the ‘good with cats’ marker rather than the ‘not good with cats’ marker.

Housetrained status showed a smaller but significant positive association with rehoming speed ($HR = 1.08, p < .001$) but by day 90, a higher percentage of dogs who were labeled not housetrained had been rehomed, perhaps because of the popularity of puppies and a high association between puppies and a ‘No’ housetraining status. The inclusion of personality traits on a dog’s profile is associated with slower rehoming speed ($HR = 0.92, p < .001$). This may seem counterintuitive. However, the *Rehome* platform does not ask users to provide personality traits when initially posting a dog’s profile; they can only be added once a profile has been published. Therefore, many dogs who are rehomed quickly likely do not have personality traits on their profiles, and personality traits may be more likely to accumulate as dogs go un-rehomed and users login at least every 30 days to reactivate and edit profiles. Description character length was not a statistically significant contributor to rehoming speed ($HR = 1.00, p = .047$).

5.6 We Very Slightly Sped up Rehoming; They Very Slightly Slowed it Down

Dogs whose pet stories contain *they* language (third-person plural), are slightly associated with slower rehoming speed ($HR = 0.96, p < .001$). By day 90, 39% of dogs whose profiles contained *they* language were rehomed, compared to 43% of dogs whose pet stories did contain it.

Word frequency analysis suggests that profiles high in *they* language may refer to pairs or groups of dogs, most likely bonded bonded pairs or litters of puppies, as shown in Table 3.

Table 3: Top words disproportionately used in profiles with *they* language

Word	% of profiles with <i>they</i> language that use the word
together	10.3%
brother	5.2%
deserve	3.2%
homes	3.0%
bonded	3.1%
separated	2.2%
pair	2.2%
separate	2.1%
girls	1.4%
sisters	1.3%

Note: % of profiles = percentage of profiles with *they* language containing the word at least once.

Randomly-selected profile excerpts using some of the top words associated with profiles high in *they* language illustrate the nature of these profiles:

*"Both have gone through training **together**, are kennel trained, know basic commands like sit and outside."* (Young Dutch Shepherd)

*"**They** are 2 years old as of June 2025 and have been with us since birth."* (Young Goldendoodle)

*"We're sadly considering rehoming Peso and possibly Duke, wonderful companions and a great **pair**."* (Senior Whippet)

The *we* LIWC category ($HR = 1.02, p = .001$) was associated with slightly faster rehoming, with dogs whose pet stories used *we* language associated with a 2% higher hazard of rehoming at any given point in time. LIWC documentation identifies the words ‘we,’ ‘our,’ ‘us,’ and ‘lets’ as examples of *we* language. Random sampling of sentences containing these *we* words demonstrate what pet stories written in the third person plural look like:

*"He came to **us** through the Waterville Humane Society as a foster, and after his early neuter, **we** couldn't bring ourselves to send him back; so **we** adopted him"* (Puppy Border Collie)

*"Please help **us** find a loving home for **our** loving dog."* (Adult Cane Corso)

*"These behavioral issues make it very difficult for **us** to keep him anymore."* (Adult Chihuahua)

The use of *we* may be a subtle indication that a dog comes from a family environment.

5.7 Time Language Has Significant but Tiny Positive Effects

Time language (e.g. new, since, date, never) was associated with a small increase in rehoming speed ($HR = 1.01, p < .001$). 44% of dogs whose pet stories included *time* language were rehomed by

day 90 compared to 40% for those with low or no prevalence. The *time* LIWC category is broad and *time* words were used in many ways across profiles—they were present in 91% of profiles. While the average profile was 561 characters in length, the average profile without *time* language was 116.7, suggesting that the time language variable may at least partially be picking up on pet stories that are shorter. We reviewed a random sample of 50 profiles without *time* language and found that they primarily listed some personality features of the dogs and contained very few other topics we identified in our qualitative content analysis. They generally failed to take a biographical position that situated dogs' stories within time, such as mentioning placement history or the dogs' future needs. For example, these are the entire pet stories of four dogs that lack *time* language:

"Hanks is an amazing family dog. He loves being outside and going on walk or roaming free. He loves kids and kids love him." (Adult Bluetick Coonhound)

"Dog is very outgoing. Loves to play with family and children. Likes going for rides." (Adult Jack Russel Terrier)

"Rarri is an sweet dog he is house-trained. He loves to play and goes for walks/runs. He is playful" (Adult Pit Bull Terrier)

"We need a good home for this sweet dog. He is full of energy but we cannot take care of him because I am allergic. He is matted and would need a groom." (Young Shi Poo)

5.8 Risk Language Has The Largest Effect on Rehoming Speed—Slowing It

Table 4: Top words disproportionately used in profiles with risk language

Word	% of profiles with risk language that use the word
protective	23.1%
safe	21.5%
alert	5.5%
secure	5.0%
safety	4.2%
prevention	4.1%
protect	3.2%
protector	3.1%
trouble	3.1%
escape	3.0%

Note: % of profiles = percentage of profiles with risk language containing the word at least once.

Risk language showed the strongest overall and most negative text effect (HR = 0.91, $p < .001$) as shown in Figure 4 and Table 5. Pet stories containing *risk*-related language were associated with slower rehoming speed. By day 90, 35% of dogs whose pet stories

containing *risk* language had been rehomed compared to 43% of those without.

19% of profiles contained some amount of *risk* language. Word frequency analysis of pet stories revealed that *risk* language reflected protective or alert behavioral descriptions of dogs, or requests by profile authors for their dogs to enter safe environments, as shown in Table 4. Reviewing randomly-selected examples of profile sentences using these words demonstrates their usage in context:

Protective

"I am a great snuggler and love to sleep with my owners to feel more secure but if someone comes by the house, I am the first one there to make sure they don't mess with my family." (Adult Great Dane)

"She not only is affectionate but also has a protective nature that will make you feel safe and secure." (Adult Labrador Retriever)

"He's a very sweet dog and has gotten protective over the kids and I think he would be protective over his new owners too." (Young Australian Cattle Dog)

Safe

"He may need a safe space where he can't mark anything important while he gets adjusted." (Adult Great Pyrenees)

"Looking for a safe family or owner for her." (Young Bichon Frise)

"This sweet girl deserves a safe, caring, and forever home where she'll continue to be loved and mentally engaged." (Young Belgian Malinois)

Alert

"She's potty trained, very alert, and loves to be near her people." (Adult Schnauzer (Miniature))

"He is alert has a beautiful soft coat." (Puppy American Staffordshire Terrier)

"He will alert you when he needs to go outside." (Young Chihuahua)

Secure

"She is a jumper and needs a secure home." (Young Mutt)

"We don't let him go near the front door and keep him in a spacious kitchen behind secure dog gates." (Adult Rhodesian Ridgeback)

"He also needs to have a home with a secure/contained backyard." (Adult Husky)

5.9 Results Summary

Overall, the three-block competing risks model achieved a concordance index of 0.731, with each successive block contributing significantly to model fit. Appearance—particularly breed and size—emerged as dominant predictors of rehoming speed for dogs on the *Rehome* platform. Responses to 'good with' categories contributed to the model, with dogs marked as good with dogs, kids, or cats rehoming faster. Linguistic features also slightly improved the model,

with *risk* language producing the most robust effects. These findings suggest that while linguistic features of pet stories contribute to rehoming speed, the effect is modest relative to appearance and compatibility characteristics of dogs captured in categorical profile features.

So why did Mocha the chihuahua wait more than four months on *Rehome*? Our results suggests some possibilities. As a small dog in the toy breed group she would be expected to be among the quickest dogs to rehome. But, Mocha is labeled with the ‘special needs’ status, although her pet story“ provides no indication of any physical special needs. She is labeled as ‘good with cats’ but ‘not good with kids’ and her ‘good with dogs’ status has been left unaddressed. ‘Good with cats’ is the least helpful of the compatibility features in increasing rehoming speed, while the most helpful—‘good with dogs’ has been left blank.

Additionally, her pet story uses more than twice as much risk language as the average pet story with risk language:

*"Mocha is a sweet and loyal 8-year-old dog who has been by my side since she was a puppy. She was given to me by my parents when I was young, and she's been my companion through everything—from hard times to happy ones. Mocha has always had a deep bond with her person and loves to be close by, offering comfort, **protection**, and affection. She is very intuitive and **protective**, especially toward the person she connects with most. Because of her strong loyalty, she can be reactive when unfamiliar people get too close to her person or come into her space. For this reason, Mocha would do best in a calm, low-traffic home with someone who has experience with **protective** or emotionally sensitive dogs."*

6 Discussion

That adopters of dogs from shelters, rescues, and on *Rehome* prioritize appearance when choosing dogs is well-established [18]. Our study confirms these characteristics strongly impact the speed of peer-to-peer rehoming on the *Rehome* platform.

Other studies have established that compatibility with other dogs, cats, and kids are also salient factors in dog adoption decisions in shelters and online [19? ?]. Our study confirms this and specifies that dog-dog and dog-kid compatibility is more associated with faster rehoming speed than dog-cat compatibility. We also confirm that dogs marked as ‘special needs’ rehome slower than those without the designation. These profile features are prominently displayed on dogs’ profiles and in site filters; owners considering rehoming should fill out as much of this information as possible.

A key contribution of our study is to demonstrate that compared to these highly salient appearance and compatibility features, the linguistic features of user-generated pet stories are less salient in peer-to-peer online dog rehoming. However, that is not to say that pet stories and their linguistic qualities do not matter in framing the “rehome-ability” of dogs on the platform. LIWC categories are intended to capture psychological cues into authors’ “thought processes, emotional states, intentions, and motivations” [?]. In pet stories, they seem to pick up on subtle uses of language that

nudge the reader toward interpretations: A kind of dog whistle for rehome-ability.

Pet stories that made more use of *time* language, perhaps suggesting a more biographical approach, were associated with faster rehoming, as were profiles that used the family-oriented *we* pronouns which might be a subtle signal that a dog comes from a family environment. However, negative linguistic signals outweighed positive ones. The most powerful linguistic contributor to rehoming speed was the use of *risk* language, which was associated with slower rehoming. This may be due to readers’ negativity bias—the tendency to give more weight to negative information than to positive—but more importantly it could be due to information asymmetry. Dog owners who provided honest disclosures of behavioral problems or personality flaws in their pet stories may have been more penalized in the marketplace relative to non-disclosers. So might have owners who were simply unaware of the dog whistles in their word choice, for example in bios with sentences like “she ... has a protective nature that will make you feel safe and secure.”

In a study of 43,517 dogs in the U.S. in which dog owners completed a survey including questions from the C-BARQ and C-BARQ mini behavioral assessment, 99.12% of dogs showed at least one moderately serious or serious behavioral problem or at least two moderate or serious problems [?]. The most common problems were the kind of separation and attachment behaviors associated with *risk* language in our study, aggression, fear, and anxiety. Because only 19% of *Rehome* pet stories in our dataset included *risk* language, it’s likely that many pet stories belonged to dogs with these common problems, but they simply weren’t referenced.

This information asymmetry is visible in the debate that has taken place in the shelter and rescue industry for many years: Whether adoptable pet stories are advertisements that should accentuate the positive or biographies that should disclose a full picture. One proposed solution has been to recognize adoption and rehoming platforms as marketplaces and to use the marketplace logic of generating interest first in and then providing full information about products. The Human Animal Support Services project by Austin Pets Alive! maintains a guide for ethical shelter pet marketing which uses this approach by distinguishing between *public* and *private* biographies. The guide encourages shelters to keep pet stories positive, descriptive, and strengths-based when viewable in public in order to generate interest, and to disclose details about pets’ specific medical, behavioral, and other issues directly with individuals who respond to these public ads [?]. Approaching pet stories as interest-generating advertisements may also be an appropriate strategy for *Rehome* users. However, *Rehome* users probably do not have the professional sophistication to manage a public/private information flow on their own. Shelters that suggest rehoming as a diversion strategy, or even rehoming platforms themselves, might create guides, user experiences, or product features that make this approach achievable by the average dog owner.

At the same time, adoption and rehoming platforms prioritize appearance-based and pre-set classifiers, which heavily outperform narrative. Categories known to be of interest to adopters like breed, age, size, and compatibility with dogs, cats, and kids are prominent in the site’s filtering infrastructure, so it is not surprising they are

among the biggest predictors of fast rehoming speed. This marketplace approach works well for desirable dogs, but it perpetuates the disadvantages seen in shelters and rescues. Pit bulls and mixed breeds dogs specifically piled up on the site, with almost one-third of these dogs in the sample still listed on the site after 90 days and one-fifth to one-quarter relinquished to shelters.

Animal welfare organizations seeking to assist their communities in pursuing self-rehoming as an alternative to shelter relinquishment might consider not only offering their professional advice or assistance to dog owners in crafting or reviewing public and private narratives and advertising dogs, but also focus specifically on the dogs who wait the longest and are rehomed the least in these efforts.

Our study has limitations, some of which pertain to the data available. While photo count was included in our analysis, photo quality or content was not assessed. Prior research has demonstrated that photo quality and content can affect adoption speed, and could be expected to play a role in rehoming speed as well [? ?]. Additionally, adoption fees were not included, as they were not available in the dataset; neither were each dog's microchip status, owner's reason for rehoming, or whether the dog was marked as 'needs an experienced adopter,' a designation that is only available to rehome users and not to shelter or rescues creating dog profiles. It's possible that *risk* language may be associated with this marker; understanding to what extent it is could help to explain how much *risk* language accurately captures owners' assessments of their dogs and how much is incidental or unintended. Additionally, we did not segment to specific groups of dogs. Identifying whether linguistic features of pet stories are more or less salient for different groups of dogs of varying popularity may reveal more granular effects of linguistic features.

7 Conclusion

The rise of online peer-to-peer pet rehoming platforms in the U.S. is a response to the persistent problem of matching pets in need of new homes with willing adopters. Dogs successfully placed through rehoming platforms avoid entering shelters and rescues and becoming one of the more than 680,000 unwanted and unowned dogs euthanized in them each year. Platforms like *Rehome* are not only a viable alternative to shelter or rescue relinquishment but also a literal life line for many dogs [?].

Yet *Rehome* and other online adoption platforms are essentially marketplaces for living products, one in which dogs have no avenue for self-advocacy. Their personalities, behaviors, and needs are mediated through platform affordances, data structures, and what users write about their dogs in pet stories. Although unchangeable, appearance-based features like age and size are most predictive of rehoming speed on the *Rehome* platform, our results show that user-made choices matter, too. For the 70% of *Rehome* users who have a month or less to rehome their dog, filling out a compatibility marker rather than skipping it and taking a biographical stance, using *we* if applicable, and avoiding overly using *risk* language when it's not warranted may be the difference between placing their dog into a new home and resorting to relinquishing them.

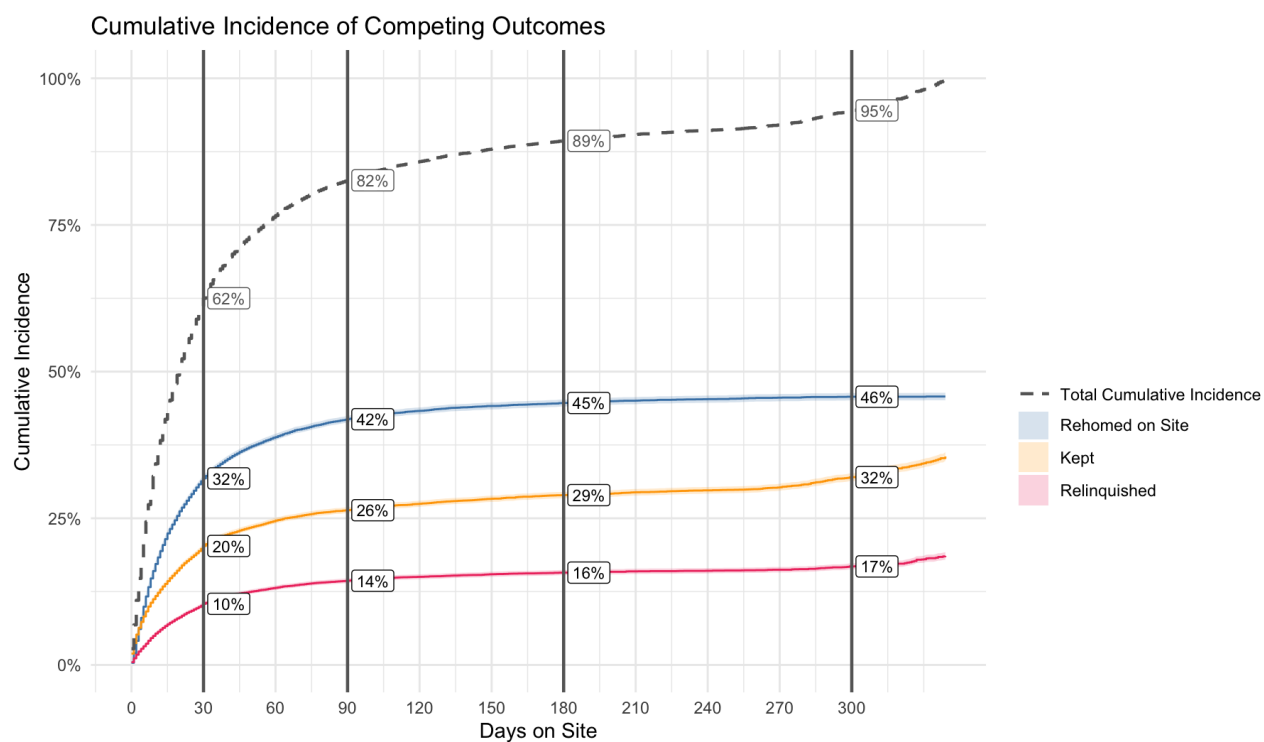


Figure 3: Cumulative incidence of competing outcomes. Most (62%) dog profiles on *Rehome* were removed within 30 days because the dog was successfully rehomed (32%), the owner decided to keep them (20%), or they were relinquished to a shelter or rescue (10%). Successful rehoming on the platform was the most common outcome by day 300 (46%). Dogs whose profiles were removed from the platform because their owner reported they were adopted off-site are not included in the analysis, but were similar in number to the amount of dogs who were kept.

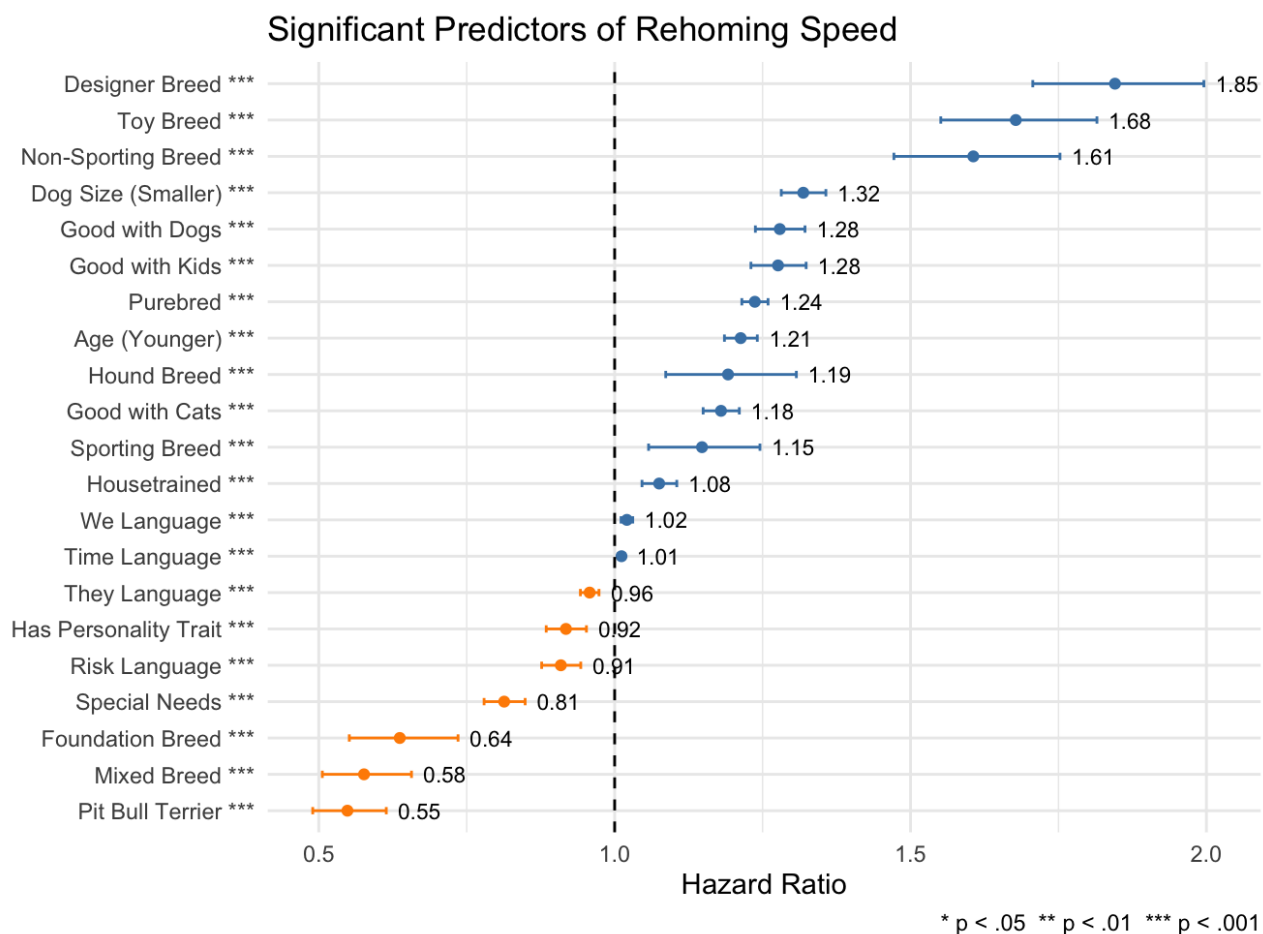


Figure 4: Profile features significantly associated with rehoming speed ($p < .001$). Breed category hazard ratios are estimated relative to Terrier Breed as the reference category, controlling for all other variables. Dogs listed as designer breeds (e.g., Goldendoodle, Labradoodle, Chiweenie) had a rehoming rate 1.86 times higher than terrier breeds. Dogs listed as Pit Bull Terriers had a rehoming rate approximately half that of terrier breeds ($HR = 0.55$). Among linguistic features derived from owner-written pet stories, risk language showed the largest effect: stories containing risk language (e.g., protective, safe, alert) were rehomed at a lower rate than those without ($HR = 0.91$).

Table 5: Cumulative Incidence of All Outcomes at Day 90 by Profile Features

Feature	Category	Cumulative Incidence at Day 90				n
		Rehomed on Site	Kept	Relinquished	Still Listed	
<i>Appearance</i>						
Age	Puppy	50%	27%	14%	9%	6,862
	Young	44%	25%	14%	17%	14,835
	Adult	38%	27%	14%	21%	14,794
	Senior	32%	31%	17%	20%	2,203
Size	Small	58%	24%	10%	8%	11,059
	Medium	36%	26%	17%	21%	16,813
	Large	33%	28%	15%	24%	9,584
	XLarge	35%	33%	12%	20%	1,238
Breed Group	Designer (e.g. Goldendoodle)	60%	26%	6.6%	7.4%	3,936
	Toy (e.g. Chihuahua)	59%	24%	9.5%	7.5%	4,966
	Non-Sporting (e.g. French Bulldog)	57%	25%	9.2%	8.8%	2,362
	Hound (e.g. Hound (Unknown Type))	42%	25%	17%	16%	2,246
	Sporting (e.g. Labrador Retriever)	40%	28%	15%	17%	4,499
	Herding (e.g. German Shepherd Dog)	37%	28%	15%	20%	6,244
	Terrier (e.g. Terrier (Unknown Type))	36%	25%	19%	20%	2,995
	Working (e.g. Husky)	35%	27%	16%	22%	5,652
	Misc (e.g. Dutch Shepherd)	31%	27%	16%	26%	140
	Foundation (e.g. American Bulldog)	25%	29%	19%	27%	1,173
	Mixed Breed	21%	28%	24%	27%	1,717
	Pit Bull Terrier	20%	27%	21%	32%	2,764
<i>Personality, Behavior, and Care</i>						
Good with Dogs	Yes	46%	25%	13%	16%	29,818
	No	22%	33%	18%	27%	4,546
	Unknown	34%	27%	19%	20%	4,330
Good with Kids	Yes	46%	25%	13%	16%	29,893
	No	26%	33%	19%	22%	2,698
	Unknown	29%	28%	21%	22%	6,103
Good with Cats	Yes	48%	25%	12%	15%	14,146
	No	27%	30%	18%	25%	7,008
	Unknown	43%	26%	15%	16%	17,540
Housetrained	Yes	42%	27%	14%	17%	32,822
	No	44%	26%	17%	13%	4,742
	Unknown	37%	23%	21%	19%	1,130
Special Needs	Yes	27%	33%	18%	22%	1,966
	No	43%	26%	14%	17%	34,718
	Unknown	35%	30%	20%	15%	2,010
Has Personality Traits	Yes	40%	26%	15%	19%	25,506
	No	45%	28%	13%	14%	13,188
<i>Pet Story LIWC Features</i>						
We Language	Present (e.g. we, our, us, let's)	42%	26%	14%	18%	18,338
	Absent	41%	27%	15%	17%	20,356
Time Language	High (e.g. new, since, date)	44%	26%	16%	18%	17,693
	Low	40%	26%	15%	19%	17,703
	Absent	40%	38%	13%	16%	3,298
They Language	Present (e.g. they, their, them, they'll)	39%	26%	16%	19%	8,516
	Absent	43%	26%	14%	17%	30,178
Risk Language	Present (e.g. protective, safe, alert)	35%	28%	16%	21%	7,288
	Absent	43%	26%	14%	17%	31,406

Note: Cumulative incidence at day 90. Rows sum to 100%. **Bold** indicates highest rehoming rate within each feature.

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